



ATILIM UNIVERSITY
Department of Computer Engineering

CMPE454 Applied Large Language Models
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Assignment Report:
Continued Pre-training, Task-Specific Fine-Tuning, and
Instruction Tuning on GPT-2
(Medical Domain)

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I. Introduction

This report is written to summarize the approach, experiments, and results of the assignment on LLMs continued pre-training and fine-tuning. The baseline and general model we aim to specialize it in the chosen medical domain, is the smallest version of GPT-2 which has 124M parameters. All the training phases of this assignment were done on the free limited version on Google Colab, which required careful configurations, as guided in the assignment guidelines, to be able to complete all the phases without crashing the memory (OOM error). Lecture notes and Colab notebooks, official documentations, technical tutorials, and AI tools were utilized to complete this assignment in the best way possible and connect what we learned in lectures throughout the semester to how to implement it as code which adheres to best practices of the field. Each section of the following represents a phase of the assignment and contains all the deliverables required with my own interpretation and learning experience.

II. Phase 0: Setting-up Google Colab

As a starting point, I set-up the files that I'll need in next stages. I loaded 6,000 abstract which equals 5.83MB of data from PubMed dataset [1], clean it, and store it in 'pretrain.txt' file. Also, for the 'classification.csv' file, I combined 250 lines of general chit-chat data (Label 0) from 'general_sentences.txt' which I created, and other 250 lines from our medical dataset to ensure equal classes (0 & 1). Lastly, I created 'instruct.json' file which contains 50 Q&A pairs generated synthetically using Gemini 3 [2]. Fig. 1. Shows the successful generation of these files.

```
Processing abstracts...
Loaded 6000 medical abstracts.

Creating pretrain.txt...
-> Success! 'pretrain.txt' created. Size: 5.83 MB
Reading 'general_sentences.txt'...
Loaded 250 general (chit-chat) sentences.

Creating classification.csv...
-> Success! 'classification.csv' created with 500 rows.

Creating instruct.json...
-> Success! 'instruct.json' created with 50 pairs.
```

Fig. 1. Output of Phase 0: Successful creation of files

III. Phase 1: The Baseline

Before I start training on the chosen medical domain data, I made initial evaluation (output quality & perplexity) of our base model (gpt2) to observe its performance on the specialized text and to compare it later after training. I loaded the base model ‘gpt2’ with the causal language modelling head (autoregressive) and its tokenizer from Hugging Face (HF) *transformers* library [3]. Then, I prompt the model (zero-shot inference) with sample text: “The idiosyncratic reaction to the drug”, the model, as shown in Fig. 2, completed the sentence in a very generic way, using plain English without using any medical terms.

--- TEST 1: Domain Specific Generation Check ---
Input Prompt: 'The idiosyncratic reaction to the drug'
Model Output:
The idiosyncratic reaction to the drug's use has been that it creates a feeling of euphoria and satisfaction.

"The more you medicate the more you feel like you're in a state where you're not feeling like a bad person,"

Fig. 2. Output of Phase 1 – Part A: The generic completion of base model

Another metric to evaluate the model was the perplexity (PPL), which can be defined as “an evaluation of the model’s ability to predict uniformly among the set of specified tokens in a corpus” [4]. A lower PPL indicates that the model is more familiar with the target corpus and can better predict the next token. In our case, using the equation and code segments from [4], I calculated the perplexity on the medical dataset and got **21.90** (Fig. 3) which is considered relatively high. As a result of this task, I learned what is PPL and how to calculate it.

To sum up, the base model lacked specialized domain knowledge and treated medical prompts as narrative or creative writing prompts.

Baseline Perplexity Score: 21.90

Fig. 3. Output of Phase 1 – Part B: The baseline PPL score

IV. Phase 2: Continued Pre-Training

After observing the baseline model behavior in Phase 1, I moved to the continued pre-training phase, using the HF *Trainer* class [5] with 3 epochs, on 6,000 PubMed abstracts [1]. Due to the limited resources, I used precision of (fp16) and a batch size of 4 with gradient accumulation steps [6], simulating an effective batch size of 16.

Although the training loss started from lower value then increased suddenly (due to initialization with random values), it demonstrated a decreasing trend through the 3 epochs as visualized in the graph below (Fig. 4)



Fig. 4. Training loss over 3 epochs showing convergence

To measure the improvement of the model in the medical domain after training, I recalculated the PPL and as expected it dropped by around 21.4% reaching **17.22**. (Fig. 5)

Specialist Perplexity Score: 17.22
Improvement: 4.68 points lower.
-> SUCCESS: The model is less surprised by medical text!

Fig. 5. The post-training PPL score

I also observed a change in the vocabulary generated by the model from narrative style to academic medical style with terms like "synthetic opiates" and "risk of relapse," which indicates that the training was successful (Fig. 6).

--- Phase 2: Domain Specific Generation Check (Specialist Model) ---
Input Prompt: 'The idiosyncratic reaction to the drug'

Specialist Model Output:
The idiosyncratic reaction to the drug is still unclear though, and one can rule out the possibility that the drugs might increase the risk of relapse. During the last few years, the development of synthetic opiates has been a significant challenge. The drug

Fig. 6. Output of Phase 2: The text generated after training adopting medical jargon

At the end I realized that even a small amount of data (< 6MB) can make a big difference in the model behavior.

V. Phase 3: Changing the Behavior of the Model (Classification Head)

In this phase, we were asked to replace the language generation head with a classification head (*AutoModelForSequenceClassification* which instantiates *GPT2ForSequenceClassification*

under the hood) [7]. The model was trained to distinguish between medical abstracts (Label 1) and general chit-chat (Label 0).

To calculate the accuracy of the classifier model, I benefited from *compute_metrics* function [8] in HF *Trainer* class. The model outputs raw prediction values (logits) for each class (0 and 1). I then applied *argmax* function to these logits to determine the class with the highest probability. Then, the predictions were compared against the ground-truth labels from the test set which represents a distinct 20% set of the complete dataset (500 rows). Finally, the total number of correct predictions is divided on the total number of evaluation samples. The model achieved a final accuracy of **99%** (Fig. 7). Apparently, the task was semantically easy, so the model can distinguish between the chit-chat and the medical abstracts data with very high accuracy.

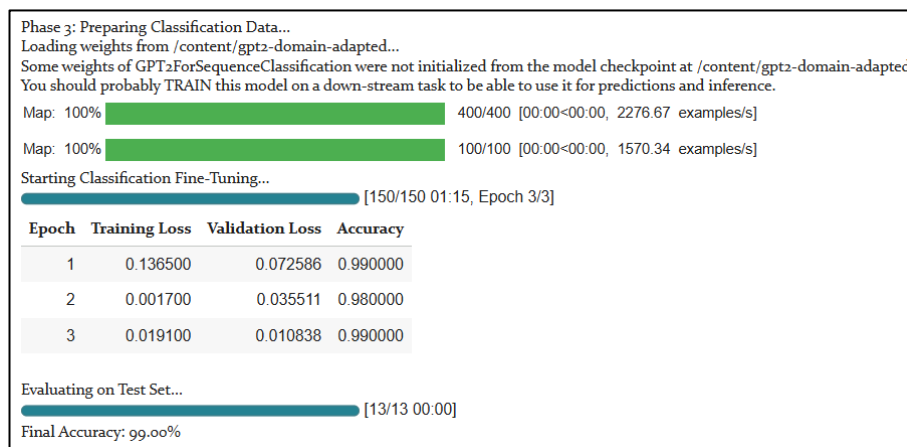


Fig. 7. The training log of Phase 3 showing the accuracy after each epoch and the final score

VI. Phase 4: Instruction Tuning & The "Abstract" Bias

To let the model act as a chatbot, I fine-tuned it on instruction pairs from ‘instruct.json’ file. An important step was to format the data strictly (an example at Fig. 8) to include “Instruction:” and “Response:” beside the ‘<EOS>’ token to teach the model when to stop generating text.

1	Instruction: What is the primary symptom of hypertension?	
2	Response: Hypertension is often asymptomatic, but severe cases may present with headaches, shortness of breath, or nosebleeds.<endofext >	
3	Instruction: Explain the function of insulin.	
4	Response: Insulin is a hormone produced by the pancreas that regulates blood glucose levels by facilitating cellular glucose uptake.<endofext >	
5	Instruction: What are the common risks of general anesthesia?	
6	Response: Risks include nausea, vomiting, sore throat, confusion, and rarely, malignant hyperthermia or cardiac complications.<endofext >	
7	Instruction: Define myocardial infarction.	
8	Response: Myocardial infarction, commonly known as a heart attack, occurs when blood flow is blocked to part of the heart muscle.<endofext >	
9	Instruction: What is the purpose of a biopsy?	
10	Response: A biopsy is performed to remove tissue or cells for laboratory analysis to help diagnose disease, especially cancer.<endofext >	
11	Instruction: What is type 2 diabetes?	
12	Response: Type 2 diabetes is a chronic condition characterized by insulin resistance and impaired blood glucose regulation.<endofext >	

Fig. 8. A sample of the formatted data, ready for instruction-tuning

I trained for 5 epochs with a lower learning rate ($1e-5$) to prevent overfitting the small dataset of only 50 Q&A pairs. The training log is shown at Fig. 9.

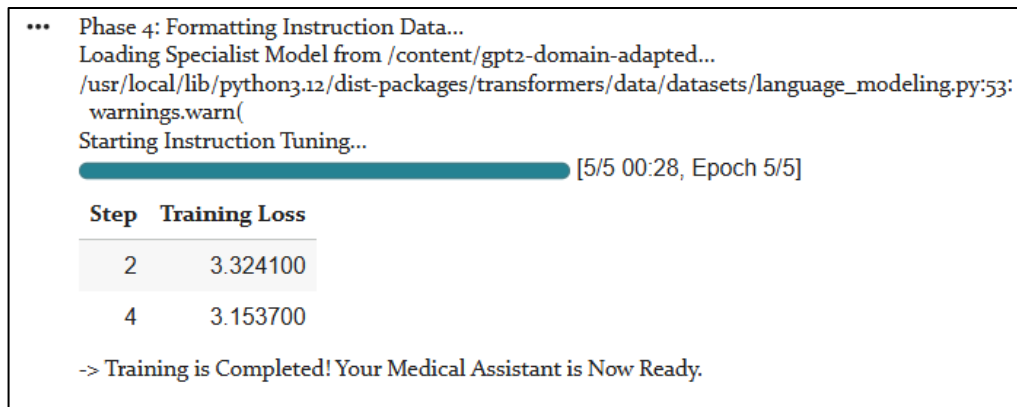


Fig. 9. Training log of Phase 4 showing loss in 2/5 steps

The model successfully adopted the Q&A format and answering medical questions directly instead of continuing the text. I tried the model by asking questions and observing its answers. I noticed that the model succeeded in adopting the academic medical tone -even though the answers seem to be not precise- but it did not learn exactly when to stop. It stops generating text, but randomly, not after a brief and complete answer. Another issue is that the model became unable to answer general questions, for example about weather, which means that catastrophic forgetting has occurred.

VII. Catastrophic Forgetting

A major observation in this assignment was the evidence of catastrophic forgetting which is the tendency of a model to lose general knowledge after specialized training. My model essentially "forgot" how to talk normally about general topics. This confirms that without using special techniques, aggressive fine-tuning on a narrow domain (like medicine) overwrites the weights responsible for general conversation. Fig. 10 demonstrates that in my interaction with the fine-tuned model, when I ask it general questions like "How is the weather today?", it starts hallucinating academic-like responses and failed to answer properly.

```

--- MEDICAL ASSISTANT TEST ---
User: What is the primary symptom of hypertension?
Bot: The most common symptom is a slight headache with fever, headache, dizziness, or nausea. Treatment is based on an understanding of the underlying pathophysiology. The patient is considered. This article reviews the literature on the role of hypertension in the development of hypertension in patients with hypertension. Most of the literature is discussed in relation to the clinical and clinical.

User: What is the function of the heart?
Bot: The heart is a central nervous system organ. Heart rate is regulated by the hypothalamus, the hypothalamus-pituitary gland, and the hypothalamus-pituitary gland. Heart rate and heart rhythm.

User: How is the weather today?
Bot: The temperature at which the temperature rises is a function of the precipitation, which in turn depends on the precipitation and on the amount of precipitation. The precipitation of precipitation.

You: Do you read books?
Bot: I did not read a few books because I was working on a computer program that used the "snowflake" algorithm. The algorithm allows us to analyze the.

You: Are you a robot?
Bot: No, I am a robot. I am a computer. I am a computer.
The next day, my son was diagnosed with leukemia. It was a difficult diagnosis to accept. I had to try to find the cure. My son was diagnosed.

You: What is a university?
Bot: The University of California, San Francisco, has a distinguished faculty of history, science, engineering, and mathematics. It is the largest university in the country and is recognized as a leader in. The purpose of this study.

You: Answer in one word, do you like pizza?
Bot: We do not like pizza. We do not like pizza.

You: exit
Exiting chat...

```

Fig. 10. My interaction with the final fine-tuned model

VIII. Conclusion

In this assignment, although the final model could not serve as a reliable medical assistant due to the limited training conditions, it successfully demonstrated the lifecycle of LLMs fine-tuning with all its benefits and restrictions. I got a practical experience in PyTorch development practices and HF high-level functions; learned new techniques to optimize memory usage like streaming in loading datasets and gradient accumulation in training; and observed how the model behavior changes across training phases and how the data amount, distribution, and quality is a key factor in model's performance. I benefited from AI tools not only to copy and paste (as their outputs are not always reliable) but as a tool that increases the efficiency and quality of my work.

Acknowledgement

AI Usage Disclosure: In accordance with the assignment guidelines, I benefited from using AI tools for coding part, and it supported me as a new learner to PyTorch framework by supporting me to complete code segments of some parts automatically. Specifically, AI was used in:

- ❖ **Code Integration:** Assist in combining code snippets which are publicly available at Hugging Face official documentation and add them to my Colab notebook.
- ❖ **Ensure Best Practices:** Debug my PyTorch code and discuss suitable training arguments to adhere to industry standards and assignment constraints (124M parameters, fp16 precision).

I acknowledge that the written content of this report is entirely my own work and reflects my interpretation, and all the codes were executed by me after I have reviewed and verified them.

References

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